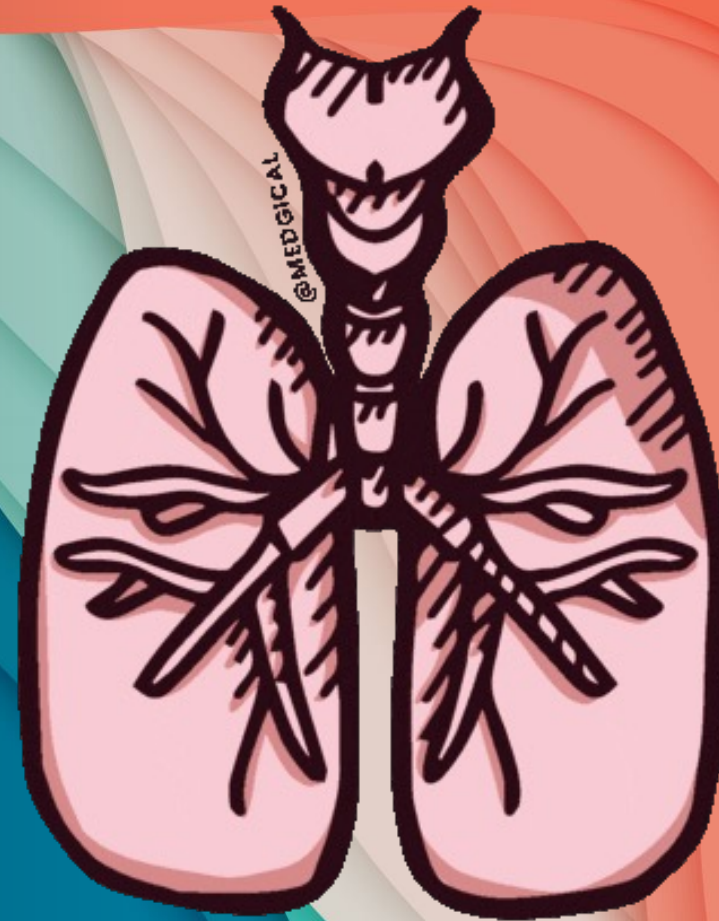
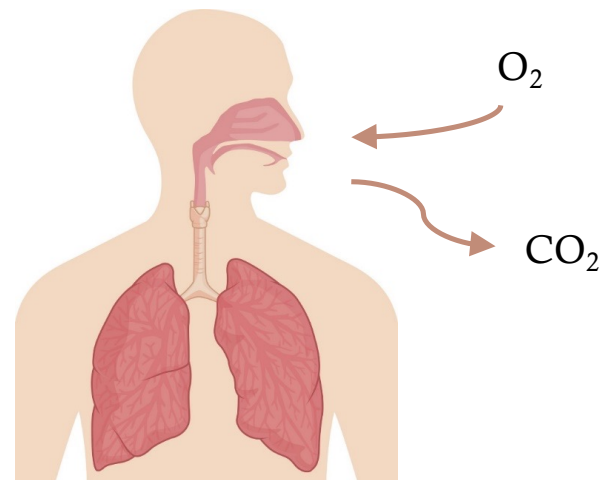

RESPIRATORY SYSTEM

Take a deep breath

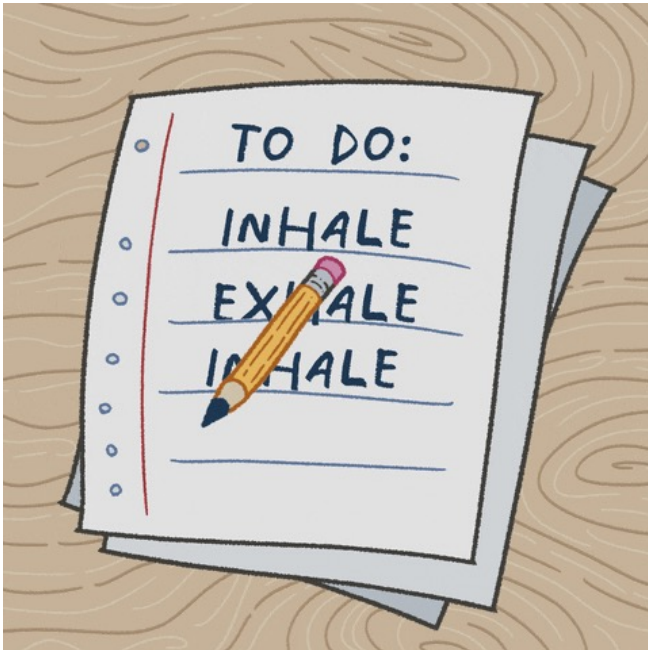


Why breathe?

- The **respiratory system** is the organ system responsible for breathing
- Our bodies take oxygen (O_2) from the air and release carbon dioxide (CO_2)
- Our cells need oxygen to perform their functions and keep us alive



HOW TO BREATHE

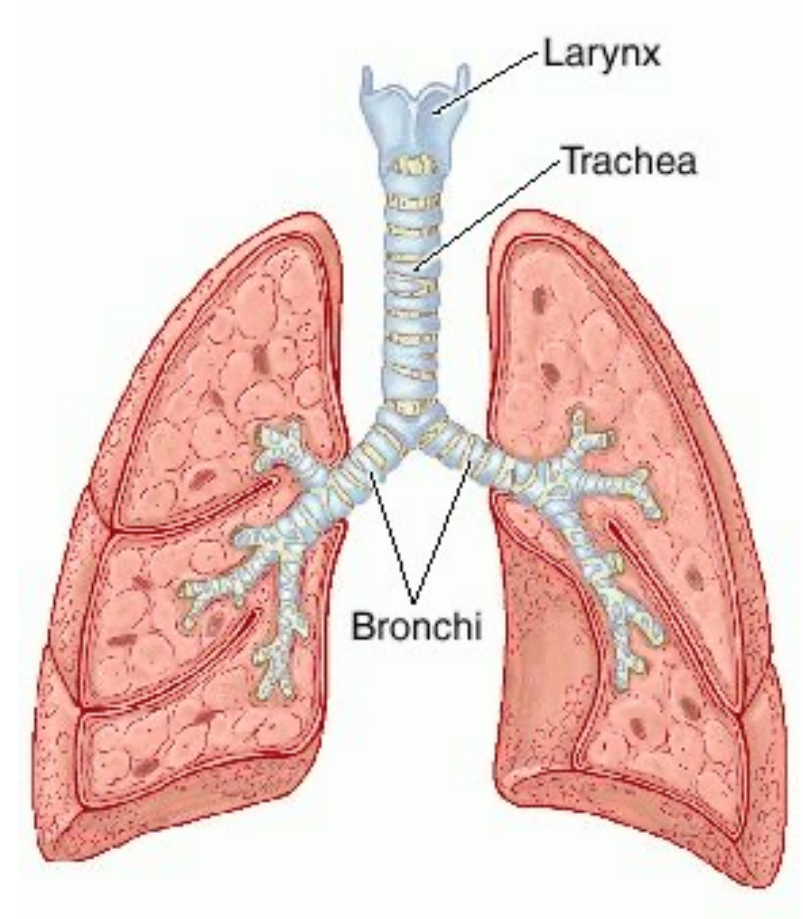


Breathing – the process of inhaling air to extract oxygen and exhaling air to release carbon dioxide

Inhalation (taking air into the lungs) occurs when the volume of the chest increases

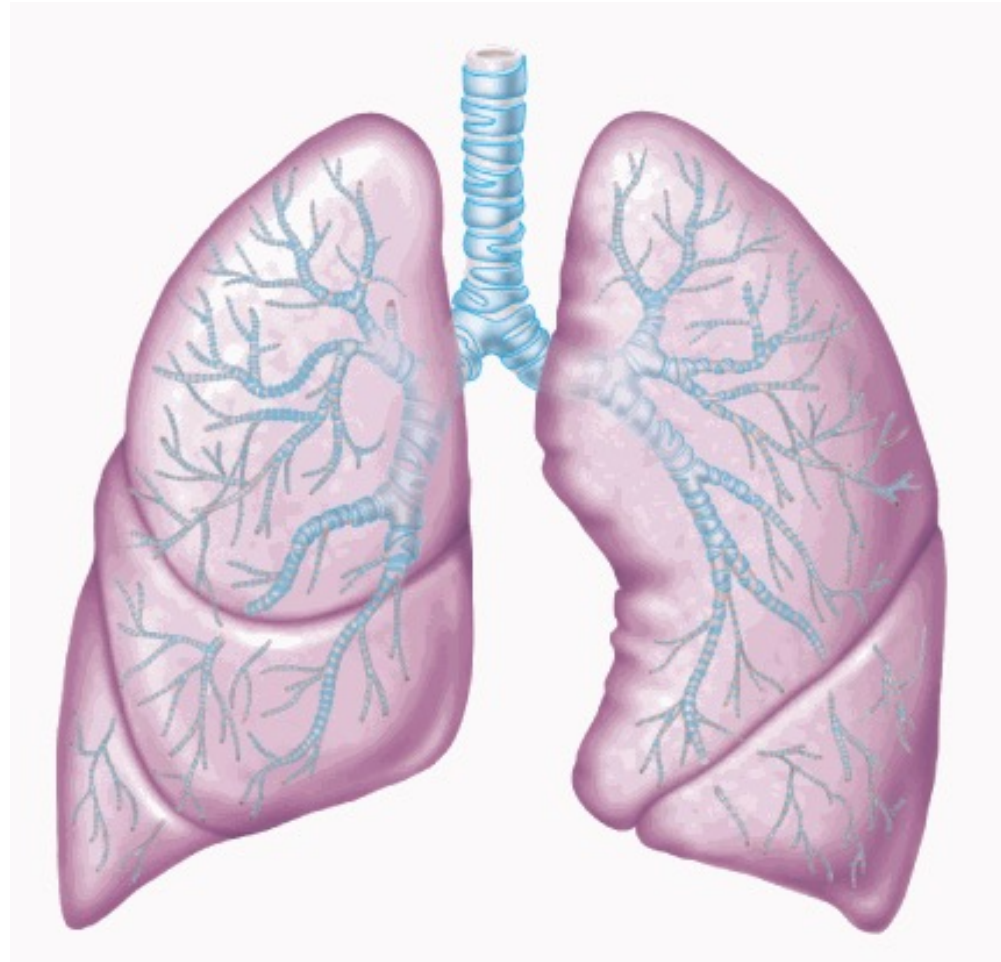
Exhalation (letting air out of the lungs) occurs when the volume of the chest decreases

-
- The **Larynx** (voice box) contains the vocal cords
 - The **Trachea** (“Windpipe”) is a large pathway for air to move down towards the lungs
 - **Bronchi** – the two main branches from the trachea to the lungs
 - The larynx, trachea, and bronchi have bands of cartilage to protect and keep them open



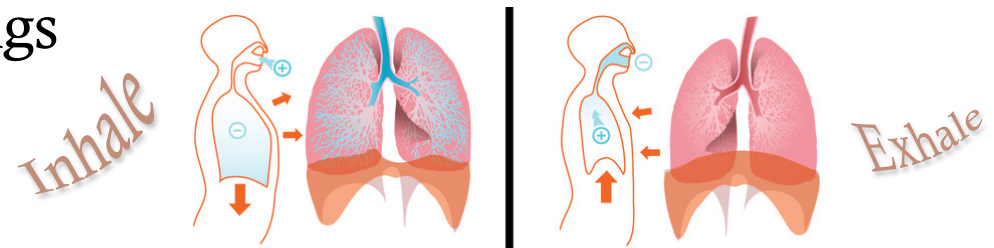
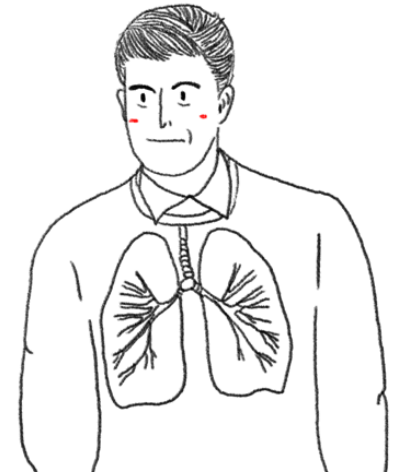
THE LUNGS

- **Lungs** – two spongy organs in the chest where the **bronchioles** are pathways to the **alveoli** where Carbon dioxide is exchanged for Oxygen
- Lungs do not contain any muscle tissue, so how does breathing happen?



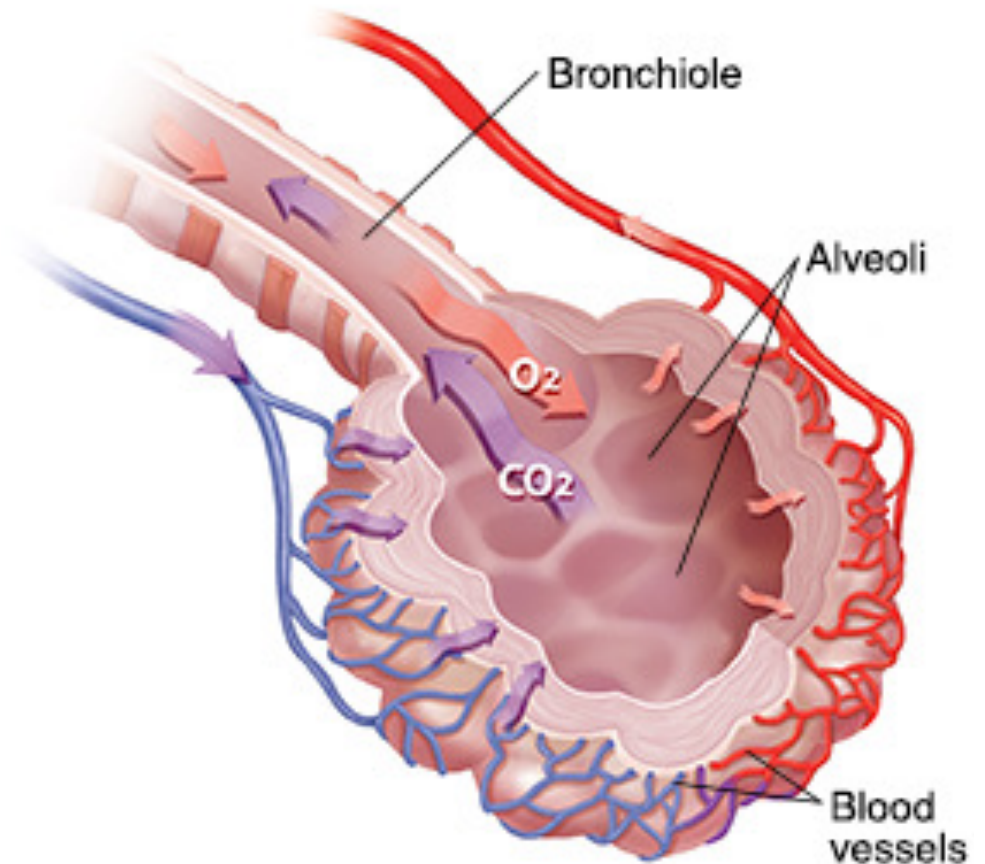
HOW TO BREATHE

- **The diaphragm** – a wall of muscle below the lungs that contracts downward
- **The rib muscles** – contract and pull the chest outward
 - When these muscles contract, the lungs fill with air
 - When these muscles relax, the chest returns to its smaller size and air exits the lungs



HOW TO BREATHE

- **Bronchioles** – Smaller tubes branching off from the bronchi
- **Alveoli** – small sacs at the ends of each bronchiole
 - Each lung has millions of alveoli
 - The alveoli are wrapped in capillaries where they transfer oxygen and carbon dioxide to the blood



WE LEARNED



Function of the
respiratory system

Organs involved in the
respiratory system

Process of breathing
